

Raspberry Pi Setup — Wireless Sensor Monitoring System

Group A, Team A6 — Advanced Embedded Systems, HSHL

Hardware Required

- Raspberry Pi 3 Model B (or B+)
 - MicroSD card (16 GB minimum, Class 10)
 - Power supply (5V / 2.5A)
 - Network connection (WiFi or Ethernet)
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Task 1 — Flash Raspberry Pi OS

1. Download **Raspberry Pi Imager** from <https://www.raspberrypi.com/software/>
2. Insert microSD card into your PC
3. In Imager: choose **Raspberry Pi OS Lite (64-bit)** (no desktop needed)
4. Click the gear icon ⚙ and configure:
 - Hostname: **raspberrypi**
 - Enable SSH → use password authentication
 - Set username: **pi** / password: **your_password**
 - Configure your WiFi SSID and password
 - Set locale / timezone
5. Write the image, insert card into Pi, power on
6. Wait ~60 seconds then SSH in:

```
ssh pi@raspberrypi.local
```

Task 2 — System Update

```
sudo apt update && sudo apt upgrade -y  
sudo apt install -y curl git
```

Task 3 — Install Mosquitto MQTT Broker

```
sudo apt install -y mosquitto mosquitto-clients

# Enable and start the service
sudo systemctl enable mosquitto
sudo systemctl start mosquitto

# Allow anonymous connections for lab testing
sudo nano /etc/mosquitto/mosquitto.conf
```

Add these lines to the config:

```
listener 1883
allow_anonymous true
```

Restart and test:

```
sudo systemctl restart mosquitto
mosquitto_sub -t "test" &
mosquitto_pub -t "test" -m "hello"
```

You should see **hello** printed. MQTT broker is working.

Task 4 — Find Pi's IP Address

```
hostname -I
```

Note the IP (e.g., **192.168.1.50**). Update the ESP32 sketch's **mqttServer** with this IP.

Task 5 — Install Node.js and Node-RED

```
# Install Node.js 18 LTS
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt install -y nodejs

# Install Node-RED globally
sudo npm install -g --unsafe-perm node-red

# Enable Node-RED as a service
sudo systemctl enable nodered.service
sudo systemctl start nodered.service
```

Verify: Open browser and navigate to <http://<Pi-IP>:1880> You should see the Node-RED editor.

Task 6 — Install Node-RED Dashboard Plugin

In the Node-RED editor:

1. Click the hamburger menu (top-right) → **Manage palette**
2. Go to the **Install** tab
3. Search for **node-red-dashboard**
4. Click **Install** → wait for completion
5. Restart Node-RED when prompted

Alternatively via SSH:

```
cd ~/.node-red
npm install node-red-dashboard
sudo systemctl restart nodered.service
```

Dashboard will be available at: <http://<Pi-IP>:1880/ui>

Task 7 — Import Node-RED Flows

Four progressive flows are provided in this folder:

File	Description
flow_01_mqtt_debug.json	MQTT receive + raw packet debug
flow_02_basic_dashboard.json	Gauge, status, knock count display
flow_03_history_charts.json	Temperature + knock history charts
flow_04_remote_control.json	Full dashboard with remote control

Import steps:

1. Open Node-RED editor at <http://<Pi-IP>:1880>
2. Hamburger menu → **Import**
3. Click **select a file to import** and choose the JSON file
4. Click **Import** → then **Deploy** (top-right red button)

Start with [flow_01](#) and verify data arrives before moving to the next.

Task 8 — Verify End-to-End

1. Power on ESP32 (ensure **mqttServer** IP matches Pi's IP)
2. Open Serial Monitor → confirm "MQTT connected"
3. On Pi terminal, run:

```
mosquitto_sub -t "sensors/node01" -v
```

You should see packets like:

```
TEMP:23.4;KNOCK:0;STATUS:NORMAL;MUTE:0;WARN:25.0;CRIT:30.0
```

4. Open <http://<Pi-IP>:1880/ui> — dashboard shows live data

Troubleshooting

Problem	Fix
ESP32 won't connect to MQTT	Check IP in sketch, verify mosquitto is running
No data in Node-RED	Check MQTT broker node topic matches sensors/node01
Dashboard blank	Ensure node-red-dashboard is installed and deployed
Pi not reachable	Use hostname -I to confirm IP, ping from PC